

Maël Madec

Robotics & Autonomous Systems Engineer

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Summary

R&D engineer in robotics and autonomous systems. Design and characterisation of autonomous systems in real-world conditions: pose estimation of underwater manipulator arms (Ifremer), real-time multi-sensor monitoring (Centrale Nantes). Python, C++, ROS2. Fluent English, C1 level (TOEIC 980), available across France & Europe.

Professional experience

Robotics Engineer Intern | September 2025 - March 2026

Ifremer - Underwater Systems Unit

- Technical study and state-of-the-art review for the modernization of the Nautile
- Kinematic and dynamic modeling of multi-axis manipulators and performance characterization in Python/MATLAB
- Development of a real-time robotics simulation environment for the Nautile and robotic arms under ROS2
- Design and development of algorithms for estimation, control, and piloting of manipulator arms on the Nautile

Assistant Engineer Intern | September 2024 - February 2025

Ecole Centrale de Nantes, GeM laboratory

- Design of a real-time monitoring system for metal additive manufacturing: multi-sensor data acquisition, processing, and fusion in Python/C++
- Sensor modeling and simulation in Python/MATLAB for characterization, correction, and calibration

Brewery Operator | July 2022 - August 2022, July 2024 - August 2024

Brasserie de Bretagne & Brasserie Poem

- Beer production, quality control, packaging, and direct sales

Education

Engineering Degree | September 2020 - June 2026

Bretagne INP - ENIB

- Multidisciplinary curriculum in computer science, electronics, mechanics, and control engineering
- Specialization in mechatronics, robotics, and embedded systems: modeling, control, and perception

Erasmus in Slovenia | February 2025 - June 2025

University of Ljubljana - Faculty of Electrotechnics

- Robotics projects: control of a UR5e arm, modeling and fuzzy control of an aerial crane
- Advanced coursework in robotics, control engineering, artificial intelligence, and autonomous systems

Erasmus in Italy | February 2022 - July 2022

Politecnico di Torino - Italy

- English-taught courses in embedded systems and networks
- Agility in a multicultural environment and fluent technical English

Skills

Robotics : control, ROS2, simulation
Control engineering: modelling, fuzzy logic, system identification
Mechatronics: STM32, electronics & instrumentation, physical measurements
Data & algorithms: machine learning, data fusion and estimation, numerical optimisation
Programming & tools: Python, C, C++, MATLAB, Git, Docker

Languages & Strengths

Languages: French: native language - English: fluent (C1 - TOEIC 980/990)
Strengths: Adaptability - Synthesis skills - Teamwork - Autonomy - Organisation